ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Low oxygen requires fast breathing.
2	**(b)** 6	$C_6H_{12}O_6 + 6\,O_2 \longrightarrow 6\,CO_2 + 6\,H_2O$.
3	**(a)** uses carbon dioxide and releases oxygen	Photosynthesis replenishes oxygen.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Valves prevent backflow.
5	**(a)** 1	$6\,CO_2 + 6\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2$.
6	**(c)** chlorophyll is needed for photosynthesis	Chlorophyll.
7	**(c)** A is true but R is false.	Bile has no enzymes; it emulsifies fats.
8	**(c)** 4320	72 x 60.
9	**(a)** both plants	No light or no CO2 means no photosynthesis.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Transpiration pull.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume per minute = 72 x 70 (1); = 5040 mL (5.04 L) (1).	1+1
12	Bile neutralises acid from the stomach (1); Pancreatic enzymes work in alkaline medium (1).	1+1
13	Low haemoglobin (1); Less oxygen to cells, less energy (1).	1+1
14	Reabsorbed 178.5 L (1); Percentage = 99.2% (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Light is needed for photosynthesis (1); Starch is formed in light only (1); Covered region shows no starch (1).	1+1+1
16	Heart works continuously (1); Needs oxygen and glucose (1); Blockage leads to heart attack (1).	1+1+1
17	Salivary amylase breaks starch to maltose (1); Stomach acid stops amylase (1); Pancreatic amylase in small intestine gives glucose (1).	1+1+1
	OR: Atmospheric oxygen is lower (1); Faster breathing to get more oxygen (1); More RBC for transport (1).	1+1+1
18	Tubule reabsorption is faulty (1); Normally all glucose is reabsorbed (1); Diabetes may be a cause (1).	1+1+1
	OR: Soil solution is more dilute (1); Water moves into root hair cell (1); Moves into xylem (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Aim: to show CO_2 is needed for photosynthesis (1); KOH absorbs CO_2 (1); Leaf with KOH: no starch (1); Leaf with water: starch present (1); Control: leaf in water flask (1).	1+1+1+1+1
	OR: Aerobic: mitochondria, CO_2 + H_2O (1); Yeast: ethanol + CO_2 (1); Muscle: lactic acid (1); Only 2 ATP in anaerobic (1); Complete oxidation yields about 30 ATP (1).	1+1+1+1+1
20	Per minute: 7500 mL (1); Per hour: 450000 mL = 450 L (1); Oxygen = 21% of 450 = 94.5 L (1); Not all oxygen is absorbed (1); Exhaled air has about 16% oxygen (1).	1+1+1+1+1
	OR: Systolic 120: ventricle contracts (1); Diastolic 80: ventricle relaxes (1); Instrument: sphygmomanometer (1); Hypertension: high pressure strains arteries (1); May cause rupture or heart disease (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 5040	70 x 72.
2	**(c)** more carbon dioxide	Exhaled air has about 4% CO_2.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Efficient oxygen supply.
4	**(c)** red blood cells	RBCs contain haemoglobin, which carries oxygen.
5	**(a)** reabsorption in nephron	Glucose is normally reabsorbed.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Short diffusion distance.
7	**(b)** 99%	(180 - 1.8)/180 = 99%.
8	**(d)** bacteria produce acid from sugar	Tooth decay.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Lactic acid accumulation.
10	**(b)** decrease in transpiration and photosynthesis rate	Gas exchange stops.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	NaOH absorbs CO_2 (1); No raw material for photosynthesis (1).	1+1
12	Phloem is cut (1); Food cannot reach the roots, roots die (1).	1+1
13	Produces food (glucose) for all organisms (1); Releases oxygen for respiration (1).	1+1
14	No gas exchange (1); No photosynthesis or transpiration (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Herbivores: long for cellulose digestion (1); Carnivores: short (1); Meat is easier to digest (1).	1+1+1
16	Evaporation increases (1); Wind removes humid air (1); Cooling and water absorption (1).	1+1+1
17	Green plant: autotrophic (1); Cuscuta: parasitic, lacks chlorophyll (1); Absorbs food from host (1).	1+1+1
	OR: Valves in heart and veins (1); Muscular contractions of heart (1); Pressure gradient arteries to veins (1).	1+1+1
18	Photosynthesis stores energy; respiration releases (1); Photosynthesis uses CO_2; respiration gives CO_2 (1); Photosynthesis only in light; respiration all the time (1).	1+1+1
	OR: Glomerulus has fine capillaries (1); High pressure filtration (1); Long tubule for reabsorption (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Renal artery to glomerulus (1); Filtered into Bowman's capsule (1); Passes through the tubule not reabsorbed (1); Collecting duct, ureter, bladder (1); Urethra (1).	1+1+1+1+1
	OR: Provides food and oxygen (1); Cactus: stomata open at night (1); Stores CO_2 as acid (1); Reduces water loss in day (1); Fixes it for photosynthesis in day (1).	1+1+1+1+1
20	Fish: two-chambered heart (1); Single circulation (1); Human: four chambers, double circulation (1); Warm-blooded need more oxygen (1); Prevents mixing (1).	1+1+1+1+1
	OR: Heart rate rises (1); Cardiac output rises (1); Breathing rate rises (1); Muscles may respire anaerobically (1); Lactic acid accumulates if oxygen supply is inadequate (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** phloem	Phloem is outside of the xylem.
2	**(a)** Both A and R are true and R is the correct explanation of A.	ATP drives loading.
3	**(b)** ethanol and carbon dioxide	Fermentation gives ethanol and CO2.
4	**(d)** away from the heart under high pressure	Arteries are away from heart.
5	**(c)** A is true but R is false.	A is true. R is false: guard cells take up water in light and swell.
6	**(c)** cytoplasm	Glycolysis occurs in the cytoplasm.
7	**(a)** urea	Urea and salts.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Replaces kidney function.
9	**(c)** 500 mL	Tidal volume.
10	**(c)** they have high energy needs	Warm blooded.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$C_6H_{12}O_6 \longrightarrow 2\,C_2H_5OH + 2\,CO_2 + \text{energy}$ (2).	2
12	It brings oxygenated blood (1); Pulmonary vein is the only vein carrying oxygenated blood (1).	1+1
13	Higher pressure (1); Oxygen supply at depth is limited (1).	1+1
14	Pulmonary: heart to lungs and back (1); Systemic: heart to body and back (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Inhalation (1); Diaphragm flattens to increase chest volume (1); Air cannot enter properly (1).	1+1+1
16	Breathing rate increases (1); Heart rate increases (1); More blood to the muscles (1).	1+1+1
17	Contraction raises ribs and flattens diaphragm (1); Chest volume increases, pressure falls (1); Air rushes in; reverse expels (1).	1+1+1
	OR: Alveoli bring oxygen near blood (1); Haemoglobin binds oxygen (1); Released in oxygen-poor tissues (1).	1+1+1
18	Urea and wastes accumulate in the blood (1); Salt and water balance is lost (1); Poisoning leads to death (1).	1+1+1
	OR: Fish have low energy needs (1); Single circulation: blood passes heart once (1); Mammals need high pressure and oxygen (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Xylem: dead hollow tubes, lignin (1); Phloem: sieve tubes with companion cells (1); Girdling removes phloem (1); Food cannot reach roots (1); Roots die, then plant (1).	1+1+1+1+1
	OR: Nostrils, pharynx, larynx, trachea (1); Bronchi, bronchioles, alveoli (1); Haemoglobin in blood (1); Heart pumps to the muscle (1); CO_2 returns to lungs in blood and is exhaled (1).	1+1+1+1+1
20	No digestion in mouth (1); Bile emulsifies fat in the intestine (1); Lipase breaks to fatty acids and glycerol (1); Absorbed by villi (1); Taken by lymph (1).	1+1+1+1+1
	OR: Urea builds up (1); Water and salt imbalance (1); Dialysis filters blood (1); Needs repeated sessions (1); Transplant is the permanent cure (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.